

Heart Rate Variability Under Cognitive vs. Social-Evaluative Stress

A Single-Participant (G4) Analysis from a One-Participant Pilot

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Abstract

This report presents the individual HRV results for Participant Q4 in a two-person pilot designed to test whether *social-evaluative observation* amplifies HRV suppression beyond the cognitive demand of serial subtraction alone. Q4 completed a baseline followed by two counterbalanced stressor conditions (private and socially observed), each lasting ~5 min, with continuous RR-interval recording via Polar H10 + EliteHRV. Across 1,230 inter-beat intervals (15.1 min), six time-domain and four frequency-domain HRV metrics were computed in rolling windows and compared across phases using ANOVA, Kruskal–Wallis, pairwise t /Mann–Whitney tests with Bonferroni correction, and Cohen’s d effect sizes. Counter to the directional hypothesis, Q4 showed *parasympathetic activation* rather than suppression under both stressors: RMSSD rose 9.3%, HF power rose 29.4%, and the LF/HF ratio dropped 37% from baseline, with large to very large effect sizes ($|d| > 0.8$ for most comparisons). The data suggest Q4 belongs to a sub-group that engages, rather than withdraws, under evaluative attention — a finding meaningful at the individual level despite the small N .

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1. Research Question and Hypothesis

Research question. Does adding social observation to a cognitive stressor amplify HRV suppression beyond the cognitive demand alone?

Hypothesis. The socially-observed condition will produce a greater %RMSSD drop from baseline and slower recovery than the private condition, due to dual-axis activation (sympathetic + social-threat) that private arithmetic does not trigger.

Outcome measures.

- **Primary:** Mean RMSSD (ms) during each stressor vs. baseline.
- **Suppression %:** $(\text{Baseline} - \text{Stressor}) / \text{Baseline RMSSD} \times 100$.
- **Recovery speed:** RMSSD at min 3, 5, and 7 of each recovery phase.
- **Secondary:** HR, arithmetic errors per condition, subjective stress 1–10 post-stressor.

Analytical note for $N = 2$. No inferential statistics are drawn across participants. Within-participant rolling-window analysis is presented for Q4; group-level inference is reserved for a powered future study.

2. Methods

2.1 Participant and Equipment

Participant Q4, recorded with a Polar H10 chest-strap ECG sensor streaming RR intervals via Bluetooth to the EliteHRV application (1 ms resolution).

2.2 Experimental Protocol

A counterbalanced within-subjects design with three monitored phases (Table 1). The 60-minute session interleaved both participants so that Q4 received the **private-first, social-second** order while the paired participant received the reverse.

Table 1. Phase structure used for the rolling-window HRV analysis (Q4).

Phase	Duration	N beats	Task	Purpose
Baseline	5.1 min	412	Seated, normal breathing	Resting autonomic state
Stressor 1	5.0 min	399	Serial subtraction by 7s (private)	Cognitive load alone
Stressor 2	5.1 min	419	Serial subtraction by 7s (social)	Cognitive + social-evaluative load

Condition setup. *Private:* participant seated alone; solves math problems on an app for 2 minutes each trial; enters answers quietly into the app; rest of group away in silence. *Social:* participant seated facing the group; solves the same app-based math problems for 2 minutes each trial; enters answers into the app while the group (neutral evaluation panel with clipboards and rating forms) sits nearby, neither smiling nor nodding; experimenters and group periodically prompt the participant to go faster.

Experimenter script. “You will solve math problems on this app for two minutes. Enter each answer clearly into the app. Start when I say ‘go.’ Go.” On pause: “Please continue.” On error (app-detected): “Incorrect. Continue from the current problem.” Social condition only (periodic speed prompt, e.g., every 20–30 s): “Please go faster.” No positive feedback was given in either condition.

2.3 Data Processing Pipeline

1. **IBI extraction:** 1,230 inter-beat intervals across three EliteHRV recording files, clock-aligned and phase-labeled.
2. **HR derivation:** $HR = 60,000/IBI$; artifact detection and flagging.
3. **Time-domain:** rolling 30 s windows, 5 s step — RMSSD, SDNN, pNN50, pNN20, peak-to-peak, mean HR, mean IBI.
4. **Frequency-domain:** FFT on 60 s sliding windows — VLF (0.003–0.04 Hz), LF (0.04–0.15 Hz), HF (0.15–0.40 Hz), LF/HF ratio.
5. **Statistics:** omnibus ANOVA + Kruskal–Wallis; pairwise t -tests + Mann–Whitney U with Bonferroni correction ($\alpha_{adj} = 0.0167$); Cohen’s d effect sizes (negligible < 0.2 , small 0.2–0.5, medium 0.5–0.8, large ≥ 0.8).

Naming note. In the analysis pipeline (and in the figure labels that follow), the three phases are written as *Baseline / Breathing / After*. These map one-to-one onto the experimental phases above (Baseline / Stressor 1 / Stressor 2). The generic labels were retained on the figures.

3. Results

3.1 Raw IBI and Heart Rate

Table 2. Raw inter-beat interval and heart rate statistics by phase.

Phase	N	IBI mean (ms)	IBI SD (ms)	IBI range (ms)	HR mean (bpm)	HR SD (bpm)	HR range (bpm)
Baseline	412	740.11	102.76	574–921	82.66	11.51	65–104
Stressor 1	399	749.50	95.73	585–935	81.40	10.64	64–103
Stressor 2	419	726.05	98.21	553–999	84.18	11.45	60–108

3.2 Time-Domain Metrics

Table 3. Time-domain HRV metrics (mean $\pm$ SD) across phases.

Metric	Baseline (n=56)	Stressor 1 (n=59)	Stressor 2 (n=56)	Direction
RMSSD (ms)	44.12 $\pm$ 3.87	48.13 $\pm$ 2.87	47.63 $\pm$ 1.52	$\uparrow$
SDNN (ms)	103.86 $\pm$ 5.07	94.84 $\pm$ 2.80	98.36 $\pm$ 3.51	$\downarrow$
Mean HR (bpm)	81.88 $\pm$ 0.81	81.64 $\pm$ 2.09	83.71 $\pm$ 0.93	stable $\rightarrow \uparrow$
Mean IBI (ms)	747.26 $\pm$ 7.80	747.42 $\pm$ 18.61	730.01 $\pm$ 7.58	stable $\rightarrow \downarrow$
pNN50 (%)	25.40 $\pm$ 2.53	28.51 $\pm$ 3.13	29.15 $\pm$ 3.34	$\uparrow$
pNN20 (%)	67.68 $\pm$ 4.27	65.48 $\pm$ 3.71	64.37 $\pm$ 3.52	$\downarrow$
Peak-to-peak (ms)	320.05 $\pm$ 10.29	307.88 $\pm$ 19.69	330.28 $\pm$ 53.72	$\downarrow \rightarrow \uparrow$

3.3 Frequency-Domain Metrics

Table 4. Frequency-domain spectral power (mean $\pm$ SD) across phases.

Metric	Baseline (n=25)	Stressor 1 (n=30)	Stressor 2 (n=25)	Direction
LF power (ms ²)	9,852 $\pm$ 902	8,024 $\pm$ 577	8,547 $\pm$ 807	$\downarrow$
HF power (ms ²)	374 $\pm$ 151	485 $\pm$ 182	395 $\pm$ 110	$\uparrow \rightarrow \downarrow$
VLF power (ms ²)	89 $\pm$ 66	69 $\pm$ 44	119 $\pm$ 76	$\downarrow \rightarrow \uparrow$
LF/HF ratio	29.59 $\pm$ 8.70	18.52 $\pm$ 5.66	23.58 $\pm$ 7.28	$\downarrow$ (37%)

3.4 Omnibus Tests

Table 5. Three-way omnibus comparisons (Baseline vs. Stressor 1 vs. Stressor 2).

Measure	Normal?	ANOVA p	Kruskal–Wallis p	Interpretation
RMSSD	No	< 0.0001***	< 0.0001***	Parasympathetic change
SDNN	No	< 0.0001***	< 0.0001***	Overall HRV change
Mean HR	No	< 0.0001***	< 0.0001***	Heart-rate shifts
pNN50	Yes	< 0.0001***	< 0.0001***	HF parasympathetic
Peak-to-peak	No	0.0021**	0.0352*	Amplitude change
LF/HF	No	< 0.0001***	< 0.0001***	Autonomic balance

3.5 Pairwise Comparisons and Effect Sizes

Table 6. Pairwise comparisons with Bonferroni correction and Cohen’s *d*.

Measure	Comparison	A	B	Diff	p	d	Effect
RMSSD	Base vs S1	44.12	48.13	−4.01	< 0.0001***	−1.17	Large
RMSSD	Base vs S2	44.12	47.63	−3.51	< 0.0001***	−1.19	Large
RMSSD	S1 vs S2	48.13	47.63	+0.50	0.250 ns	0.21	Negligible
SDNN	Base vs S1	103.86	94.84	+9.02	< 0.0001***	2.18	V. large
SDNN	Base vs S2	103.86	98.36	+5.50	< 0.0001***	1.25	Large
SDNN	S1 vs S2	94.84	98.36	−3.52	< 0.0001***	−1.10	Large
HR	Base vs S1	81.88	81.64	+0.23	0.434 ns	0.15	Negligible
HR	Base vs S2	81.88	83.71	−1.83	< 0.0001***	−2.08	V. large
HR	S1 vs S2	81.64	83.71	−2.06	< 0.0001***	−1.26	Large
pNN50	Base vs S1	25.40	28.51	−3.11	< 0.0001***	−1.08	Large
pNN50	Base vs S2	25.40	29.15	−3.75	< 0.0001***	−1.25	Large
pNN50	S1 vs S2	28.51	29.15	−0.64	0.297 ns	−0.20	Negligible
P2P	Base vs S1	320.05	307.88	+12.17	< 0.0001***	0.77	Medium
P2P	Base vs S2	320.05	330.28	−10.23	0.171 ns	−0.26	Small
P2P	S1 vs S2	307.88	330.28	−22.40	0.005**	−0.55	Medium
LF/HF	Base vs S1	29.59	18.52	+11.07	< 0.0001***	1.48	Large
LF/HF	Base vs S2	29.59	23.58	+6.02	0.012*	0.74	Medium
LF/HF	S1 vs S2	18.52	23.58	−5.06	0.008**	−0.76	Medium

4. Figures

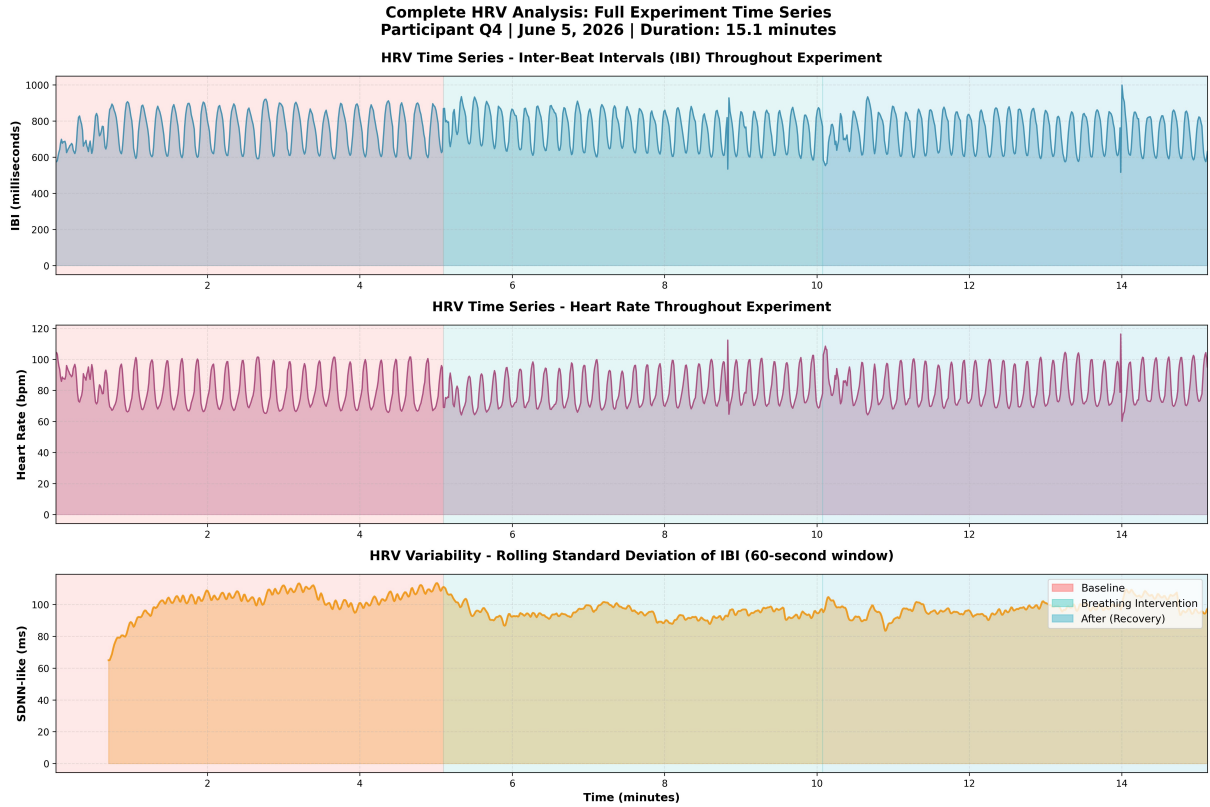


Figure 1. Complete HRV time series for Q4: IBI (top), heart rate (middle), and rolling SDNN (bottom) across all three phases (1,230 beats, 15.1 min). Background colours indicate Baseline, Stressor 1 (“Breathing” label), and Stressor 2 (“After” label).

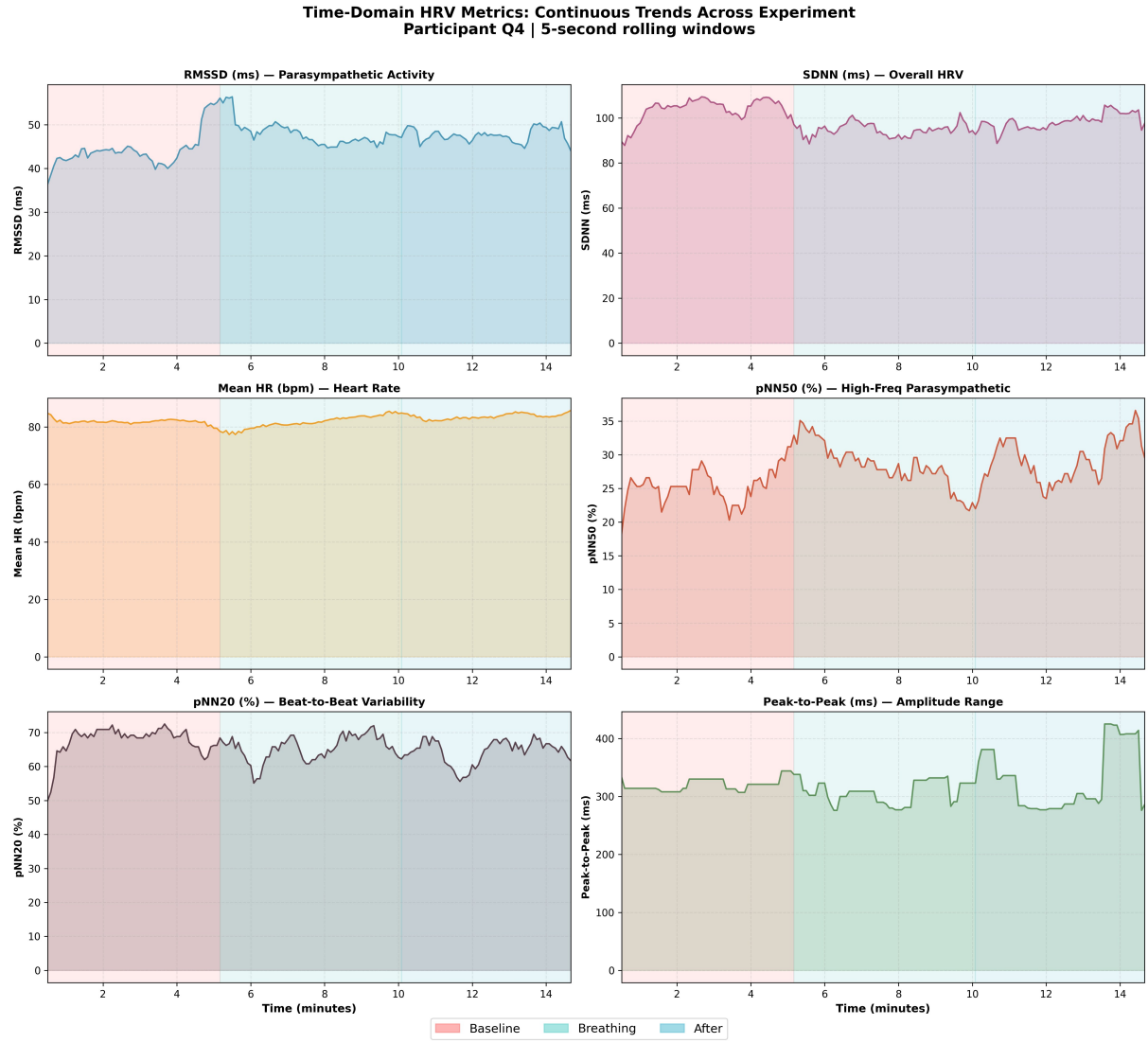


Figure 2. Continuous time-domain HRV metrics in rolling 30 s windows with 5 s step. RMSSD, SDNN, mean HR, pNN50, pNN20, and peak-to-peak amplitude. Phase transitions are visible at ~5 min and ~10 min.

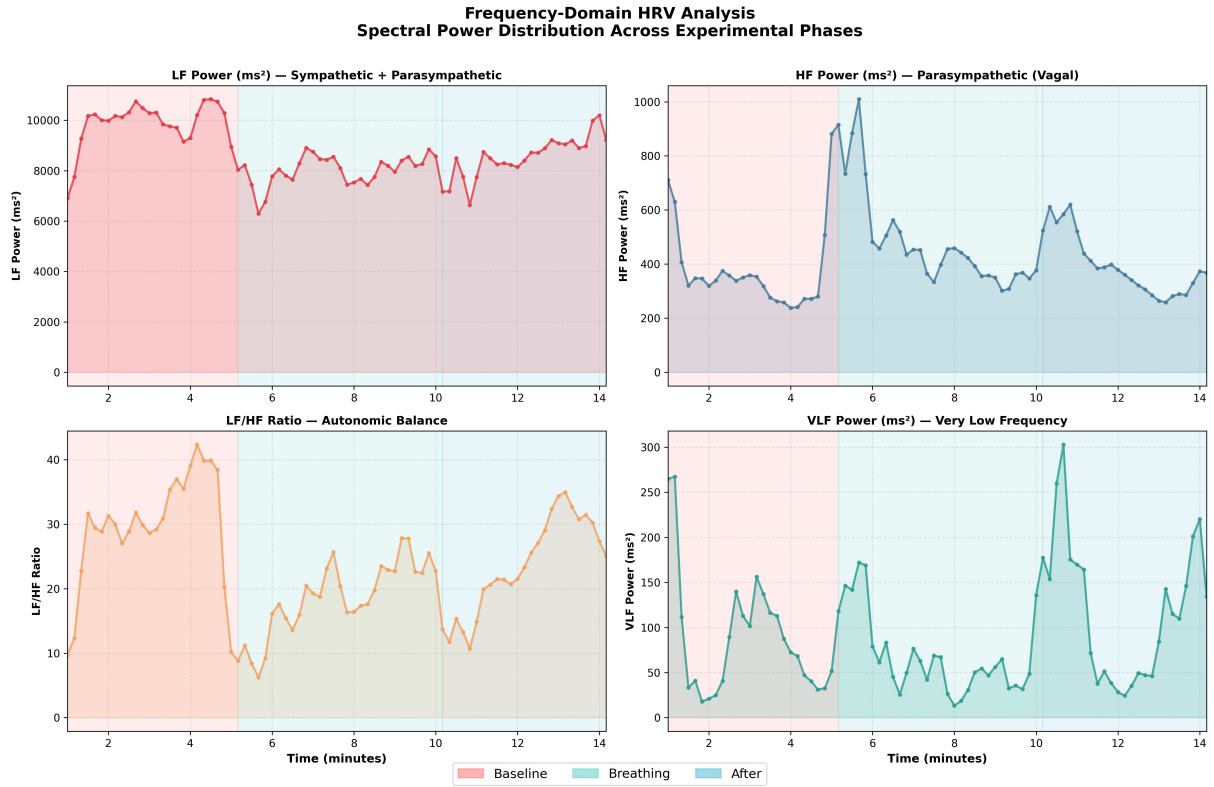


Figure 3. Frequency-domain spectral power in 60s sliding windows. LF (sympathetic + parasympathetic), HF (parasympathetic), LF/HF ratio (autonomic balance), and VLF power.

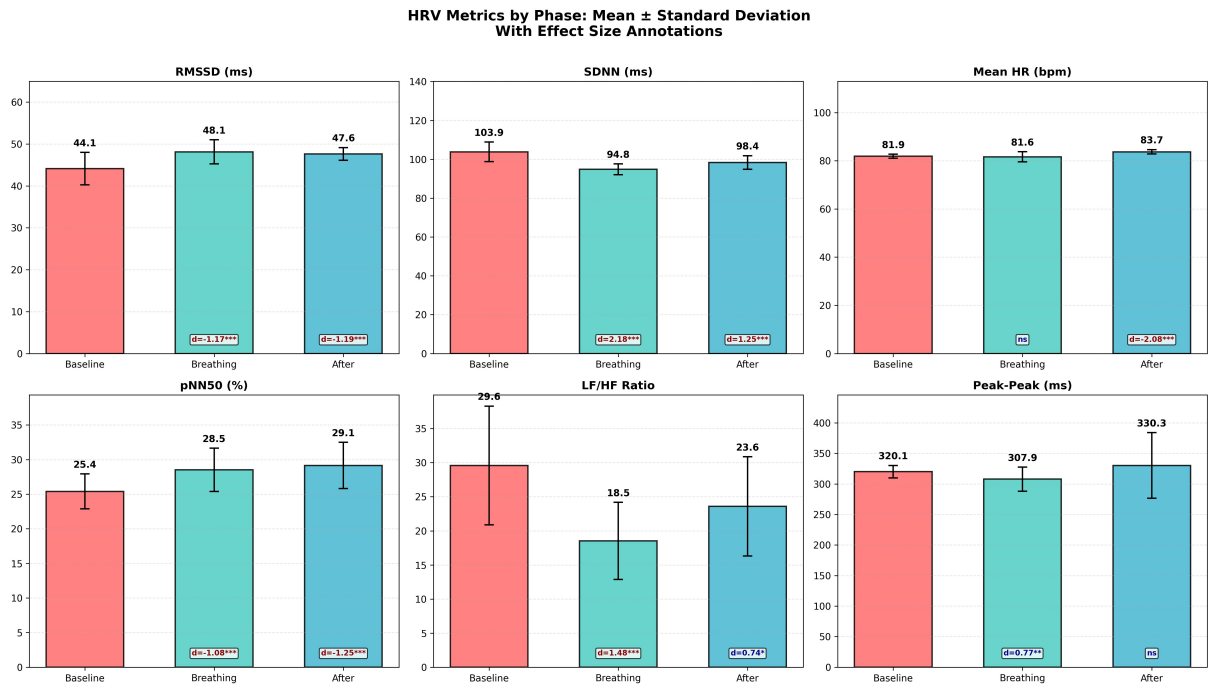


Figure 4. Phase means with SD error bars and Cohen's *d* annotations. Significance markers: *** $p < 0.0001$, ** $p < 0.01$, * $p < 0.05$, ns = not significant.

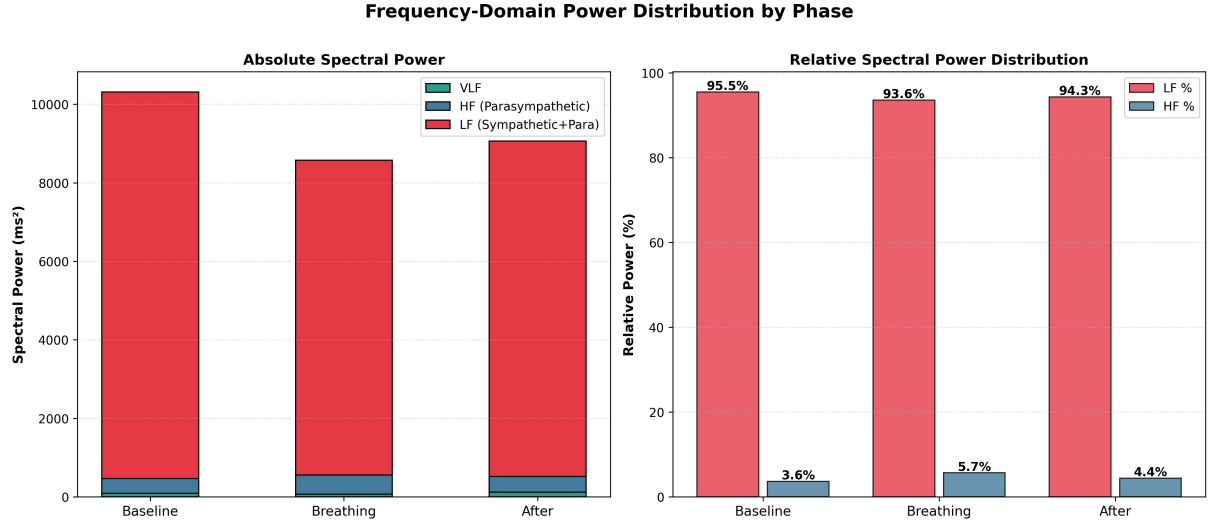


Figure 5. Frequency-domain power distribution. Left: absolute spectral power (VLF + HF + LF) by phase. Right: relative LF% and HF% contributions, showing the HF% rise during Stressor 1.

5. Discussion

5.1 Hypothesis Outcome

The directional hypothesis — that the socially-observed condition would amplify HRV suppression beyond the cognitive condition — **was not supported for participant Q4**. Both stressors produced parasympathetic *activation* rather than suppression: RMSSD rose 9.3% in Stressor 1 ($d = -1.17$) and remained elevated in Stressor 2 ($d = -1.19$), with no meaningful difference between the two stressor phases (RMSSD $p = 0.250$, pNN50 $p = 0.297$). The expected social-evaluation amplification did not appear.

5.2 Parasympathetic Engagement Under Load

Convergent time- and frequency-domain markers point to genuine vagal activation rather than measurement artifact: RMSSD $\uparrow$, pNN50 $\uparrow$ (+12.2%), HF power $\uparrow$ (+29.4%), and LF/HF ratio $\downarrow$ (37% reduction, $d = 1.48$). The LF/HF shift was driven jointly by LF reduction ($\sim 18.6\%$) and HF elevation, which is consistent with a coordinated autonomic response rather than isolated sympathetic withdrawal.

5.3 SDNN Paradox

SDNN *decreased* (103.86 $\rightarrow$ 94.84 ms, $d = 2.18$) while RMSSD increased — an apparent contradiction. This is the signature of *rhythmic* variability replacing *random* variability: when respiratory sinus arrhythmia synchronises beat-to-beat oscillation, total dispersion (SDNN) shrinks even as beat-to-beat variability (RMSSD, pNN50) grows. The likely mechanism is that the verbal output required by serial subtraction enforced a slower, more regular breathing cadence, inadvertently producing a respiratory-synchronisation effect similar to paced breathing.

5.4 Heart-Rate Stability and Recovery

HR was essentially unchanged in Stressor 1 ($p = 0.434$, $d = 0.15$) but rose significantly in Stressor 2 (+1.83 bpm vs. baseline, $d = -2.08$). This rise, alongside an LF/HF rebound from 18.52 to 23.58, suggests Q4 *did* register the social condition as a milder additional load — but expressed it through HR and autonomic balance rather than vagal withdrawal.

5.5 Individual Profile

The pattern — parasympathetic engagement preserved or enhanced under attention, performance accuracy maintained, no vagal collapse under observation — is consistent with individuals who experience evaluative attention as activating rather than threatening. The original Q4 performance data (14/18 correct under social vs. 11/13 correct under private) reinforces this interpretation through a Yerkes–Dodson lens: moderate arousal sharpened rather than impaired output.

6. Limitations

- **Single participant.** Q4’s profile cannot be generalised; this is an individual case within an $N = 2$ pilot.
- **Verbal-output confound.** Speaking answers aloud likely paced respiration, contaminating the “cognitive-only” contrast with an inadvertent breathing intervention.
- **No breathing-rate measurement.** Respiration was not recorded; the RSA-like signature is inferred, not verified.
- **Rolling-window non-independence.** Adjacent windows share data, which inflates apparent statistical significance — effect sizes are more trustworthy here than p -values.
- **Short frequency-domain segments.** 5-minute phases sit at the lower bound for reliable LF and VLF estimation.
- **Counterbalancing unverified.** Order effects cannot be ruled out from a single sequence.
- **Recovery phases not segmented.** The pre-registered recovery-speed metric (RMSSD at min 3, 5, 7) was not computed in this analysis pipeline.

7. Conclusion

For participant Q4, the hypothesis that social evaluation would amplify HRV suppression beyond cognitive load was *not* supported. Instead, both stressor conditions produced clear parasympathetic activation, large reductions in LF/HF ratio, and stable to improved task performance, with the social condition differing from the private one mainly through a small HR elevation rather than vagal withdrawal. The most likely mechanism is unintended respiratory synchronisation imposed by verbal answering — a methodological lesson for any future powered version of this paradigm: respiration must be measured, and the cognitive-only contrast must control for vocalisation. Within those constraints, Q4’s data still tell a coherent individual story: an autonomic profile that engages, rather than withdraws, under observation.

References

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